

critical thinking questions

1. As you may know, alcohol is considered a depressant drug, yet often individuals do not act depressed while under the influence. How does neuroscience explain this? What is the difference between a central nervous system depressant and a mood depressant? While under the influence of alcohol, how is behavior affected when the following brain structures are depressed: medulla, cerebellum, frontal lobe (impulse control), Broca's area, and the hippocampus?
2. Should sports (like football) that have high incidences of concussions be banned especially for teens and children since we now know the long-term negative neural impact of these injuries?
3. Laboratory research with rats has shown that dendrites develop more complex and intricate branching when subjects are exposed to enriched and variable external stimulation than when raised in more impoverished environments. What might this mean for children who are reared in deprived or limited environments? What should we do to enable individuals to develop their potential?
4. The nature of neural messages is both electrical and chemical. Explain the electrical nature of the action potential and the chemical nature of communication across the synapse.
5. A frail elderly woman had managed to save enough money for a large and very heavy flat screen television. When a fire breaks out in her apartment, she is somehow able to push the television out of her apartment. After the fire is extinguished, it takes two very strong firemen to push the television back into the apartment. How would you account for the woman's ability to push the television out? In your answer, be certain to reference the endocrine system and the autonomic nervous system.